Обратная сторона Луны

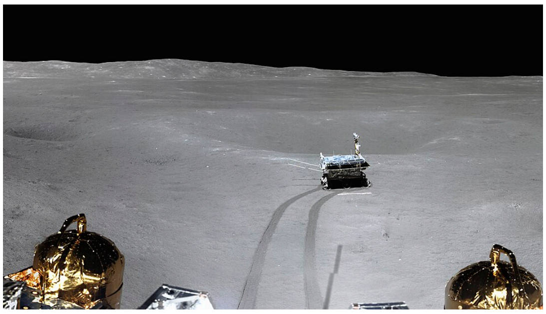
Дальняя сторона Луны, также называемая **темной стороной Луны**, - это полушарие Луны, обращенное в сторону от Земли; противоположным полушарием является ближняя сторона. Из-за блокировки приливов время, необходимое для того, чтобы Луна один раз обошла вокруг Земли, равно времени, необходимому для того, чтобы Луна один раз повернулась, таким образом, дальняя сторона Луны никогда полностью не видна с поверхности Земли. Дальнюю сторону иногда называют «тёмной стороной Луны», где «тёмная» означает «невидимая», а не «неосвещённая». Однако существует распространённое заблуждение, что тёмная сторона Луны называется так потому, что на неё никогда не попадает свет, хотя в каждом месте на Луне две недели царит солнечный свет, а в противоположном месте — ночь.[1][2][3][4] На самом деле обратная сторона более отражающая, чем видимая, поскольку на ней нет больших участков более тёмной марии.[5]

Фотография обратной стороны Луны, на которой видны Море Восточное (в центре слева) и *море кратера Аполлон* (вверху слева). Снимок сделан космическим аппаратом «Орион» во время миссии «Артемида-1».

Из-за колебаний и либрации с Земли иногда можно увидеть около 18% обратной стороны Луны. Остальные 82% оставались невидимыми до 1959 года, когда их сфотографировал советский космический аппарат «Луна-3».[6]Академия наук СССР опубликовала первый атлас обратной стороны Луны в 1960 году. Астронавты «Аполлона-8» стали первыми людьми, которые лично увидели обратную сторону Луны, когда в декабре 1968 года вышли на лунную орбиту. По сравнению с видимой стороной, обратная сторона имеет неровный рельеф с множеством ударных кратеров и относительно небольшим количеством плоских и темных лунных морей («океанов»), что делает ее похожей на другие бесплодные места Солнечной системы, такие как Меркурий и Каллисто. Здесь находится один из крупнейших кратеров в Солнечной системе — бассейн Южный полюс — Эйткен.

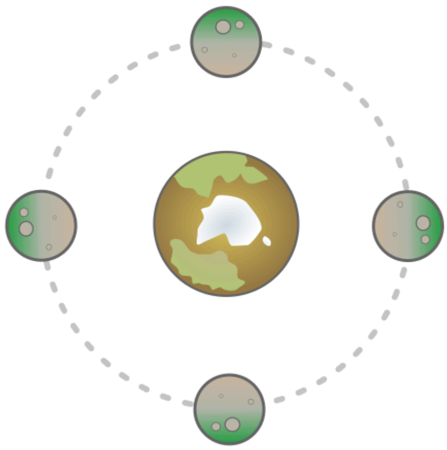
Все мягкие посадки с экипажем и без него происходили на ближней стороне Луны до 3 января 2019 года, когда космический аппарат «Чанъэ-4» совершил первую посадку на обратной стороне.[7][8]«Чанъэ-6» миссия по возвращению образцов была запущена 3 мая 2024 года, совершила посадку в бассейне Эйткен в южном полушарии обратной стороны Луны и вернулась на Землю месяц спустя, 25 июня, с первыми в истории человечества лунными образцами, доставленными с обратной стороны. [9][10][11]

Астрономы предложили установить большой радиотелескоп на обратной стороне Луны, где она будет защищать его от возможных радиопомех с Земли.[12]



Поверхность обратной стороны Луны с марсоходом *Юйту-2* (в центре), снятая посадочным модулем Чанъэ-4.

Определение



Due to tidal locking, the inhabitants of the central body (Earth) will never be able to see the green area of the satellite (Moon).

Приливные силы со стороны Земли замедлили вращение Луны до такой степени, что одна и та же сторона всегда обращена к Земле — это явление называется приливным захватом. Поэтому другая сторона, большая часть которой никогда не видна с Земли, называется «обратной стороной Луны». Со временем некоторые края обратной стороны, имеющие форму полумесяца, становятся видимыми из-за либрации.[13] В общей сложности с Земли в разное время можно увидеть 59 процентов поверхности Луны. Полезное наблюдение за участками обратной стороны Луны, которые иногда видны с Земли, затруднено из-за низкого угла обзора с Земли (их нельзя наблюдать «в полный рост»).

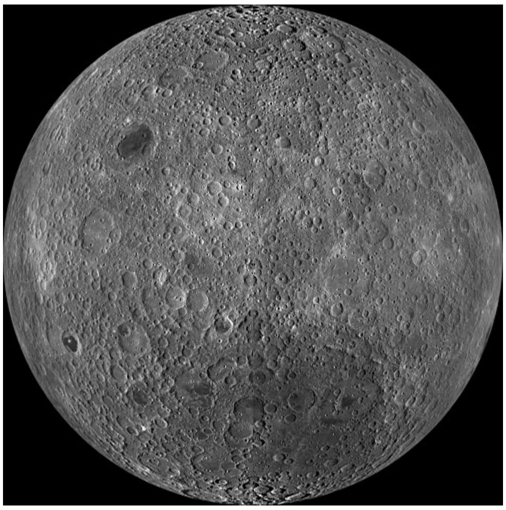
A common misconception holds that the Moon does not rotate on its axis. If that were so, the whole of the Moon would be visible to Earth over the course of its orbit. Instead, its rotation period matches its orbital period, meaning it turns around once for every orbit it makes: in Earth terms, it could be said that its day and its year have the same length (i.e., ~29.5 earth days).

Despite a common misconception, the phrase "dark side of the Moon" does not refer to "dark" as in the absence of light, but rather "dark" as in unseen: until humans were able to send spacecraft around the Moon, this area had never been seen.[1][2][3][6] In reality, both the near and far sides receive, on average, almost equal amounts of light directly from the Sun. This symmetry is complicated by sunlight reflected from the Earth onto the near side (earthshine),[14] and by lunar eclipses, which occur only when the far side is already dark. Lunar eclipses mean that the side facing Earth receives fractionally less sunlight than the far side when considered over a long period of time.

At night under a "full Earth" the near side of the Moon receives on the order of 10 lux of illumination (about what a city sidewalk under streetlights gets; this is 34 times more light than is received on Earth under a full Moon) whereas the far side of the Moon during the lunar night receives only about 0.001 lux of starlight.[14] Only during a full Moon (as viewed from Earth) is the whole far side of the Moon dark.

The word *dark* has expanded to refer also to the fact that communication with spacecraft can be blocked while the spacecraft is on the far side of the Moon, during Apollo space missions for example.[15]

Различия



Детальный снимок с Морем Москвы, сделанный лунным орбитальным аппаратом (LRO)

Два полушария Луны сильно отличаются друг от друга. Ближняя сторона покрыта множеством больших *морей* (в переводе с латыни — «моря», поскольку первые астрономы ошибочно полагали, что эти равнины — это моря лунной воды).

Обратная сторона Луны испещрена кратерами и имеет неровную поверхность с небольшим количеством морей. Морями покрыта лишь 1% поверхности обратной стороны,[16] в то время как на ближней стороне этот показатель составляет 31,2%, что делает обратную сторону более светлой.[5] Одно из общепринятых объяснений этой разницы связано с более высокой концентрацией тепловыделяющих элементов в полушарии ближней стороны, что было доказано с помощью геохимических карт, полученных с помощью гамма-спектрометра «Лунар Проспектор». Хотя другие факторы, такие как высота над уровнем моря и толщина коры, также могли влиять на места извержения базальтов, они не объясняют, почему обратная сторона бассейн Южный полюс — Эйткен (где находятся самые низкие точки Луны и где кора тоньше) не была столь вулканически активной, как Океан Бурь на видимой стороне.

It has also been proposed that the differences between the two hemispheres may have been caused by a collision with a smaller companion moon that also originated from the Theia collision.[17] In this model, the impact led to an accretionary pile rather than a crater, contributing a hemispheric layer of extent and thickness that may be consistent with the dimensions of the far side highlands. The chemical composition of the far side is inconsistent with this model.

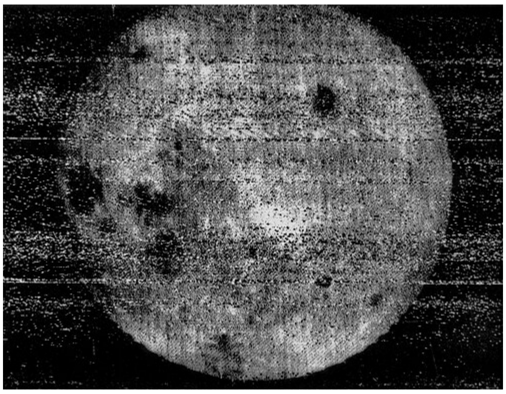
The far side has more visible craters. This is thought to be a result of the effects of lunar lava flows, which cover and obscure craters, rather than a shielding effect from the Earth. NASA calculates that the Earth obscures only about 4 square degrees out of 41,000 square degrees of the sky as seen from the Moon. "This makes the Earth negligible as a shield for the Moon [and] it is likely that each side of the Moon has received equal numbers of impacts, but the resurfacing by lava results in fewer craters visible on the near side than the far side, even though both sides have received the same number of impacts."[18]

Newer research suggests that heat from Earth at the time when the Moon was formed is the reason the near side has fewer impact craters. The lunar crust consists primarily of plagioclases formed when aluminium and calcium condensed and combined with silicates in the mantle. The cooler far side experienced condensation of these elements sooner and so formed a thicker crust; meteoroid impacts on the near side would sometimes penetrate the thinner crust here and release basaltic lava that created the maria, but would rarely do so on the far side.[19]

The far side exhibits more extreme variations in terrain elevation than the near side. The Moon's highest and lowest points, along with its tallest mountains measured from base to peak, are all located on the far side.[20]

Исследование

Ранние исследования



Снимок, сделанный 7 октября 1959 года аппаратом «Луна-3», на котором впервые видна обратная сторона Луны. Хорошо видны Море Москвы (вверху справа) и тройка морей Море Кризисов, Море Дождей и Море Смита (в центре слева).

До конца 1950-х годов о невидимой стороне Луны было мало что известно. Либрации периодически позволяли мельком увидеть объекты у лунного лимба на невидимой стороне, но это была лишь часть общей поверхности Луны — до 59 %. [21] Эти объекты были видны под низким углом, что затрудняло их изучение (было сложно отличить кратер от горного хребта). Остальные 82% поверхности на обратной стороне оставались неизвестными, и их свойства были предметом многочисленных предположений.

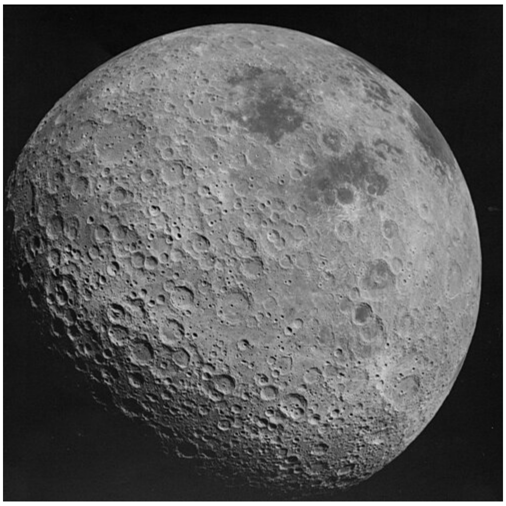
An example of a far side feature that can be seen through libration is the Mare Orientale, which is a prominent impact basin spanning almost 1,000 km (600 miles), yet this was not even named as a feature until 1906, by Julius Franz in *Der Mond*. The true nature of the basin was discovered in the 1960s when rectified images were projected onto a globe. The basin was photographed in fine detail by Lunar Orbiter 4 in 1967. Before space exploration began, astronomers expected that the far side would be similar to the side visible to Earth.[22]

On 7 October 1959, the Soviet probe Luna 3 took the first photographs of the lunar far side, eighteen of them resolvable,[23][22] covering one-third of the surface invisible from the Earth.[24] The images were analysed, and the first atlas of the far side of the Moon was published by the USSR Academy of Sciences on 6 November 1960.[25][26] It included a catalog of 500 distinguished features of the landscape.[27] In 1961, the first globe (1:13 600 000 scale)[28] containing lunar features invisible from the Earth was released in the USSR, based on images from Luna 3.[29]

On 20 July 1965, another Soviet probe, Zond 3, transmitted 25 pictures of very good quality of the lunar far side,[30] with much better resolution than those from Luna 3. In particular, they revealed chains of craters, hundreds of kilometers in length,[24] but, unexpectedly, no mare plains like those visible from Earth with the naked eye.[22] In 1967, the second part of the *Atlas of the Far Side of the Moon* was published in Moscow,[31][32] based on data from Zond 3, with the catalog now including 4,000 newly discovered features of the lunar far side landscape.[24] In the same year, the first *Complete Map of the Moon* (1:5 000 000 scale[28]) and updated complete globe (1:10 000 000 scale), featuring 95 percent of the lunar surface,[28] were released in the Soviet Union.[33][34]

As many prominent landscape features of the far side were discovered by Soviet space probes, Soviet scientists selected names for them. This caused some controversy, though the Soviet Academy of Sciences selected many non-Soviet names, including Jules Verne, Marie Curie and Thomas Edison.[35] The International Astronomical Union later accepted many of the names.

Further survey missions



The far side of the Moon, with Mare Marginis and Mare Smythii visible, photographed by Apollo 16 in 1972. It is much more cratered than the near side of the Moon.

On 26 April 1962, NASA's Ranger 4 space probe became the first spacecraft to impact the far side of the Moon, although it failed to return any scientific data before impact.[36]

The first truly comprehensive and detailed mapping survey of the far side was undertaken by the American uncrewed Lunar Orbiter program launched by NASA from 1966 to 1967. Most of the coverage of the far side was provided by the final probe in the series, Lunar Orbiter 5.

The far side was first seen directly by human eyes during the Apollo 8 mission in December, 1968. Astronaut William Anders described the view:

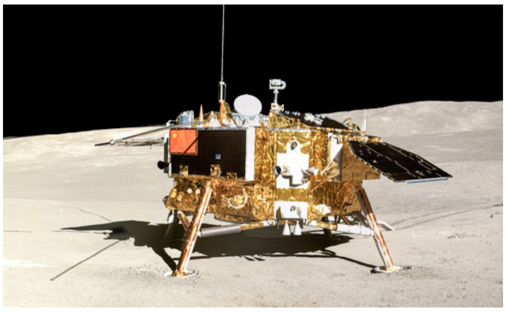
> "The backside looks like a sand pile my kids have played in for some time. It's all beat up, no definition, just a lot of bumps and holes."

It has been seen directly by all 24 men who flew on Apollo 8 and Apollo 10 through Apollo 17 and the three men and one woman on Artemis II, and photographed by multiple lunar probes. Spacecraft passing behind the Moon were out of direct radio communication with the Earth, and had to wait until the orbit allowed transmission. During the Apollo missions, the main engine of the Service Module was fired when the vessel was behind the Moon, producing some tense moments in Mission Control before the craft reappeared.

Geologist-astronaut Harrison Schmitt, who became the last to step onto the Moon, had aggressively lobbied for Apollo 17's landing site to be on the far side of the Moon, targeting the lava-filled crater Tsiolkovskiy. Schmitt's ambitious proposal included a special communications satellite based on the existing TIROS satellites to be launched into a Farquhar–Lissajous halo orbit around the L2 point so as to maintain line-of-sight contact with the astronauts during their powered descent and lunar surface operations. NASA administrators rejected these plans on the grounds of added risk and lack of funding.

The idea of utilizing the Earth–Moon L_2 for a communications satellite covering the Moon's far side has been realized, as China National Space Administration launched the Queqiao relay satellite in 2018.[37] It has since been used for communications between the Chang'e 4 lander and *Yutu-2* rover, which successfully landed in early 2019 on the lunar far side, and ground stations on the Earth. L2 is proposed to be "an ideal location" for a propellant depot as part of the proposed depot-based space transportation architecture.[38]

Soft landing



The Chang'e-4 lander imaged by the *Yutu-2* rover on the lunar far side.

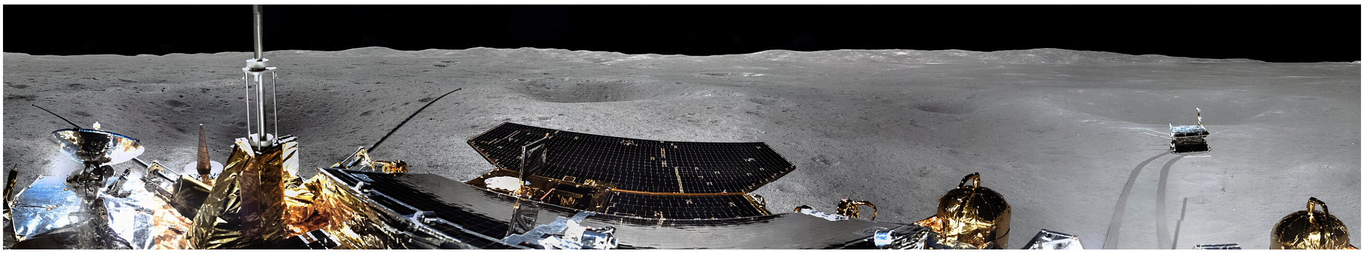
The China National Space Administration (CNSA)'s Chang'e 4 achieved the first ever soft landing on the lunar far side on 3 January 2019 and deployed the *Yutu-2* lunar rover onto the lunar surface.[39]

The craft included a lander equipped with a low-frequency radio spectrograph and geological research tools.[40] The far side of the Moon provides a good environment for radio astronomy as interferences from the Earth are blocked by the Moon.[41]

In February 2020, Chinese astronomers reported, for the first time, a high-resolution image of a lunar ejecta sequence, as well as direct analysis of its internal architecture. These were based on observations made by the Lunar Penetrating Radar (LPR) on board the *Yutu-2* rover.[42][43]

CNSA launched Chang'e 6 on 3 May 2024, which conducted the first lunar sample return from Apollo Basin on the far side of the Moon.[44] It was CNSA's second lunar sample return mission, the first achieved by Chang'e 5 from the lunar near side four years earlier.[45] It also carried a mini "Jinchan" rover to conduct infrared spectroscopy of lunar surface and imaged the Chang'e 6's lander on the lunar surface.[46] The lander-ascender-rover combination was separated with the orbiter and returner before landing on 1 June 2024 at 22:23 UTC. It landed on the Moon's surface on 1 June 2024.[47][48] The ascender was launched back to lunar orbit on 3 June 2024 at 23:38 UTC, carrying samples collected by the lander, and later completed another robotic rendezvous and docking in lunar orbit. The sample container was then transferred to the returner, which landed in Inner Mongolia on 25 June 2024, completing China's far side sample return mission.[49]

The Lunar Surface Electromagnetics Experiment (LuSEE-Night) lander, a mission to soft land as early as 2026 a robotic observatory on the far side designed to measure electromagnetic waves from the early history of the universe is being developed by NASA and the United States Department of Energy.[50]



The first panorama from the far side of the Moon taken by Chang'e 4

Возможные варианты использования и миссии

Поскольку обратная сторона Луны защищена от радиоизлучения с Земли, она считается подходящим местом для размещения радиотелескопов, которыми будут пользоваться астрономы. Небольшие чашеобразные кратеры представляют собой естественную основу для стационарного телескопа, подобного Аресибо в Пуэрто-Рико. Для телескопов гораздо большего размера кратер Дедал диаметром 100 километров (60 миль) расположен недалеко от центра обратной стороны Луны, а его 3-километровый (2 мили) край помог бы блокировать помехи от орбитальных спутников. Еще один потенциальный кандидат для размещения радиотелескопа — кратер Саха.[51]

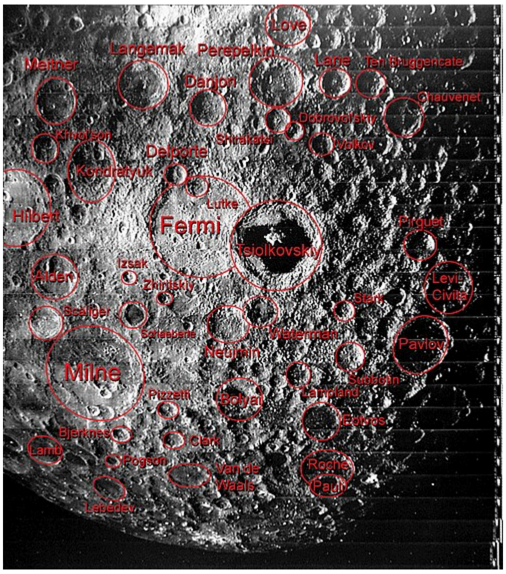
Прежде чем развернуть радиотелескопы на обратной стороне Луны, необходимо решить несколько проблем. Мелкая лунная пыль может загрязнить оборудование, транспортные средства и скафандры. Проводящие материалы, используемые для радиотарелок, также должны быть тщательно защищены от воздействия солнечных вспышек. Наконец, территория вокруг телескопов должна быть защищена от помех, создаваемых другими радиоисточниками.

The L_2 Lagrange point of the Earth–Moon system is located about 62,800 km (39,000 mi) above the far side, which has also been proposed as a location for a future radio telescope which would perform a Lissajous orbit about the Lagrangian point.

One of the NASA missions to the Moon under study would send a sample-return lander to the South Pole–Aitken basin, the location of a major impact event that created a formation nearly 2,400 km (1,500 mi) across. The force of this impact has created a deep penetration into the lunar surface, and a sample returned from this site could be analyzed for information concerning the interior of the Moon.[52]

Because the near side is partly shielded from the solar wind by the Earth, the far side maria are expected to have the highest concentration of helium-3 on the surface of the Moon.[53] This isotope is relatively rare on the Earth, but has good potential for use as a fuel in fusion reactors. Proponents of lunar settlement have cited the presence of this material as a reason for developing a Moon base.[54]

Названные функции



Some of the features of the geography of the far side of the Moon are labeled in this image

- Эйткен (кратер)
- Амичи (кратер)
- Анучин (кратер)
- Аполлон (кратер)
- Авогадро (кратер)
- Белькович (кратер)
- Белопольский (кратер)
- Бергстранд (кратер)
- Беркнер (кратер)
- Биркхофф (кратер)
- Бьеркнес (лунный кратер)
- Бок (лунный кратер)
- Кэмпбелл (лунный кратер)
- Кантор (кратер)
- Карно (кратер)
- Кэрролл (кратер)
- Кассегрен (кратер)
- Чендлер (кратер)
- Чаппелл (кратер)
- Чернышев (кратер)
- Комри (кратер)
- Бассейн Кулона-Сартона
- Крукс (кратер)
- д'Аламбер (кратер)
- Дедал (кратер)
- Дэвиссон (кратер)

- Дебус (кратер)
- Дельпорт (кратер)
- Дайсон (кратер)
- Эллерман (кратер)
- Эмден (кратер)
- Эно-Пельтери (кратер)
- Ферми (кратер)
- Финсен (кратер)
- Флеминг (кратер)
- Фаулер (кратер)
- Фридман (кратер)
- Ганский (кратер)
- Герасимович (кратер)
- Гуллстранд (кратер)
- Хейн (кратер)
- Хегу (кратер)[55][56]
- Герцшпрунг (кратер)
- Герберт Уэллс (кратер)
- Гиппократ (лунный кратер)
- Узо (кратер)
- Икар (кратер)
- Целостность (кратер)
- Иоффе (кратер)
- Изсак (кратер)
- Дженнер (кратер)
- Камерлинг-Оннес (кратер)
- Кирквуд (кратер)
- Клют (кратер)
- Kolhörster (crater)
- Комаров (кратер)
- Королев (лунный кратер)
- Ковалевская (кратер)
- Красовский (кратер)
- Куглер (кратер)
- Кулик (кратер)
- Ягненок (кратер)
- Lacus Luxuriae
- Lacus Oblivionis
- Посадочный модуль (кратер)
- Ланжевен (кратер)
- Лебедев (кратер)
- Leibnitz (crater)
- Лукреций (кратер)
- Lunar south pole
- Maksutov (crater)
- McKellar (crater)
- Mare Australe
- Mare Frigoris
- Mare Humboldtianum
- Mare Ingenii
- Mare Moscoviense
- Mare Orientale
- Mendeleev (crater)
- Michelson (crater)
- Montes Cordillera
- Montes Rook
- Mons Tai[55][56]
- Nicholson (lunar crater)

- Nishina (crater)
- Ohm (crater)
- Oppenheimer (crater)
- Oresme (crater)
- Pannekoek (crater)
- Paraskevopoulos (crater)
- Parenago (crater)
- Patsaev (crater)
- Perrine (crater)
- Pettit (lunar crater)
- Pirquet (crater)
- Pogson (crater)
- Priestley (lunar crater)
- Quetelet (crater)
- Rowland (crater)
- Sarton (crater)
- Schlesinger (crater)
- Shaler (crater)
- Shternberg (crater)
- Shuleykin (crater)
- Sikorsky (crater)
- Sniadecki (crater)
- Sommerfeld (crater)
- South Pole–Aitken basin
- Statio Tianhe[55][56] (Chang'e 4 landing site)
- Stebbins (crater)
- Stoletov (crater)
- Sverdrup (crater)
- Tianjin (crater)[55][56]
- Tikhov (lunar crater)
- Titov (crater)
- Tsander (crater)
- Tsinger (crater)
- Tsiolkovskiy (crater)
- Tyndall (lunar crater)
- Vallis Bouvard
- Vallis Inghirami
- van't Hoff (crater)
- Van de Graaff (crater)
- Van der Waals (crater)
- Vavilov (crater)
- Vertregt (crater)
- Virtanen (crater)
- Volkov (crater)
- Von Kármán (lunar crater)
- Von Neumann (crater)
- Von Zeipel (crater)
- Wan-Hoo (crater)
- Wiener (crater)
- Wright (lunar crater)
- Yamamoto (crater)
- Zhinyu (crater)[55][56]

Обратная сторона Земли

Хотя Земля не находится в приливном захвате с Луной и поэтому не всегда обращена к ней одной и той же стороной, у Земли есть «дальняя сторона» по отношению к Луне, на которой образуется «дальняя сторона» приливная выпуклость, обращенная к Луне.[57]

См. также

- Геология Луны
- Гипотеза гигантского столкновения
- Видимая сторона Луны

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Внешние ссылки

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- NASA впервые снимает на видео тёмную сторону Луны (https://web.archive.org/web/20120203002658/http://www.foxnews.com/scitech/2012/02/02/nasa-probe-captures-1st-video-moons-far-side/)

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